A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Aquaculture - Set 1

Mark scheme

AQA spec 3.5.2.3; Biological resources; textbook pp. 352-359

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 MEDCs:** market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) **1 LEDCs:** herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs – warm climates suit tilapia **1** Herbivores are more energy-efficient (lower trophic level) so more food per unit area **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm **1** Crossed with normal females (XX) all offspring are XX females **1** Female trout meat has a better flavour – in tilapia the reverse (all males) because males grow larger / more efficiently **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 4 points below. **1 Salmon need high oxygen:** water sprays / weirs aerate – wastes removed to prevent deoxygenation **1** Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density **1** Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish – controls smoltification **1** Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **deoxygenation** → loss of dissolved oxygen from the water	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
02.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Extensive:** low inputs of energy, food and chemicals so low cost and low environmental impact – sustainable – but low yield per area and dependent on natural productivity **up to 2 Intensive:** control of all limiting factors gives very high yield and profit per area – but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes **up to 2** Judgement depends on species, market and site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4

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Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation **1** Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily – fry fed fishmeal pellets and moved through tanks as they grow **1** After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection **1** Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose **1** Eyestalk ablation stimulates breeding – high densities increase disease so antibiotics / pesticides used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	

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Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2** Carnivores fed pellets of sandeels, herring, anchovy – overfishing sandeels reduces puffins and terns – reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) **2** High density spreads lice (which spread to wild fish) and infections – pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife – antibiotics breed resistant bacteria – reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow **2** Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors – reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) **2** Faeces and uneaten food decompose causing deoxygenation under cages – reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	8
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Predators are killed under licence to protect stock, reducing wild populations of protected / valued species **1** Better cage designs, netting, fencing, bird scarers exclude predators without killing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish **1** Species occupy different niches / trophic levels so they do not compete and all food sources are used **1** Bottom feeders re-suspend nutrients boosting plankton / plant growth **1** Predators control over-breeding species (tilapia) so remaining fish grow larger – total yield exceeds any single species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
04.2	**1 mark each** 1 mark for each of the 4 points below. **1 IMTA:** fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) **1** Wastes of one species become nutrients for another so less pollution and extra harvests **1 Aquaponics:** aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) **1** Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish) – rice-fish paddies similar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages **1** Production still depends on (and removes juveniles from) the wild population – carnivores need large fish inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
05.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9

Do not write outside the box

Question	Marking guidance	Marks
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	

Do not write outside the box

Question	Marking guidance	Marks
05.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 2 400 / 2 000 **1** = 1.2 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • A bare wrong answer with no working.	2
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** Carnivorous species have a high market value, so they are the most profitable to farm, and consumers in wealthy countries prefer them **1** They are fed on fishmeal and fish oil made from wild-caught fish, so farming them still depends on catching wild fish and does not take the pressure off marine stocks **1** Several tonnes of wild fish may be needed per tonne of salmon, so the extra trophic level means the total amount of fish protein available falls rather than rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

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Question	Marking guidance	Marks
06.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Uneaten feed and faeces fall through the cages and accumulate on the sea bed, where their decomposition removes oxygen and smothers the benthic community **1** The nutrients released cause local eutrophication and can trigger algal blooms **1** Parasites, especially sea lice, build up in the dense stock and spread to wild salmon and sea trout passing the cages **1** Antibiotics, pesticides used to treat sea lice and antifouling chemicals from the nets disperse into the water and affect non-target species **1** Escaped farmed fish interbreed with wild populations, diluting the locally adapted gene pool, and compete with them for food and spawning sites **1** Predators such as seals and cormorants attracted to the cages are shot or trapped **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
06.4	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Integrated multi-trophic aquaculture:** seaweeds and filter feeders such as mussels are grown alongside the cages and take up the dissolved nutrients and particulate waste as their own food **1** Siting cages where the current is strong, and leaving sites fallow between cycles, so wastes are dispersed and the sea bed recovers **1** Lowering stocking density and vaccinating the fish, so less medication is needed, and using cleaner fish such as wrasse to remove sea lice instead of pesticides **1** Stronger, better maintained nets and predator exclusion nets, so fewer fish escape and predators do not have to be killed **1** Replacing part of the fishmeal in the feed with plant or insect protein, reducing the demand on wild fish stocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

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Question	Marking guidance	Marks
06.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Farming carnivorous species still depends on wild fish for feed, so expanding it increases rather than reduces the catch of wild fish **1** Suitable sheltered coastal sites are limited, and they compete with other uses and with sensitive habitats such as mangroves, which shrimp farming has already destroyed on a large scale **1** Some valuable species, such as tuna, cannot yet be bred reliably in captivity, so farms are stocked with wild-caught young fish and the wild population is still exploited **1** The local pollution, disease and escape problems mean that simply scaling the industry up multiplies its impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-359; AQA spec 3.5.2.3.	

Do not write outside the box